Mahmoud Alyosify

Data Scientist

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SUMMARY

Data scientist who builds models around **the decision they need to support**. Predicted satellite collision risk 48+ hours ahead, measured review effort on AI-written code, and designed evaluations with proper statistics. MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30); Microsoft Certified: Data Analyst Associate; six years teaching probability and statistics.

SKILLS

Analysis: statistics, hypothesis testing, experiment design, feature engineering, imbalanced data
Modelling: scikit-learn, XGBoost, LightGBM, PyTorch
Data: SQL, Pandas, NumPy, PySpark
Reporting: Power BI, Matplotlib, Streamlit
Programming: Python, R, SQL

PROJECTS

SCRAP — satellite collision risk prediction (Python, scikit-learn) 2026

- Predicted satellite collision risk using only data available **48+ hours** before closest approach, so the warning arrives in time to act.
- Engineered features from raw conjunction records and handled a highly imbalanced target with metrics suited to rare, costly events.

Human vs. AI-Agent Pull Requests — research study (Python, GitHub data mining)

- Built a reproducible pipeline that mines GitHub to compare review effort on pull requests written by AI agents and by people.

SimCLR-Vision-SSL — self-supervised image model (PyTorch, ONNX, FAISS) 2026

- Ran a controlled 42-experiment study to find which image augmentations actually improve the model; best result 84.30% top-1 accuracy.

EXPERIENCE

AI Engineer Intern — Injaz Tech (Injaz Group), Cairo Oct 2026 – Present

- Designed the evaluation: retrieval benchmarks (top-1, MRR), category-level scores and an LLM judge checked against a fixed metric (agreement 0.82).

Systems Engineer (Military Service) — National Company for Roads Building & Development, Cairo Feb 2024 – Mar 2025

- Co-developed an internal data-management and reporting application in **C#, .NET, Entity Framework** and **SQL Server**.

Part-Time Machine Learning Instructor — National Telecommunication Institute (NTI), Egypt Jul 2025 – Present

- Teach **90-hour machine learning bootcamps** and AI for Business modules in NTI's Summer Internships Program.

Online Instructor — Udemy & YouTube (Arabic courses), Remote 2020 – Present

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

EDUCATION

MSc in Artificial Intelligence — Queen's University, Kingston, Canada Sep 2025 – Oct 2026

- GPA **4.30 / 4.30** · Completed October 2026 · Digilians Presidential Scholarship

BSc in Computer & Information Science (Bioinformatics) — Assiut University, Egypt Sep 2019 – Jul 2023

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

CERTIFICATIONS & AWARDS

Microsoft Certified: Data Analyst Associate · IBM AI Developer Professional Certificate (IBM, 2026) · Deep Learning Specialization (DeepLearning.AI, 2026) · Machine Learning Specialization (DeepLearning.AI) · 1st Place, Smart Cities Hackathon 2022 (77 university teams)

LANGUAGES

Arabic (native) · English (professional working proficiency)